MTH1004

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2019

Probability, Statistics and Data

Module Leader: Dr Daniel Williamson

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Graph Paper and Statistical Tables (Murdoch & Barnes) will be provided. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) There are twenty nine stalls at the Farmer's Market. All sell cheese, meat or vegetables or a combination of these. Fourteen stalls sell meat, eighteen sell cheese and twenty vegetables. Ten stalls sell meat and vegetables, eight sell cheese and meat and eleven sell cheese and vegetables. How many stalls sell all three (cheese, meat and vegetables)? (8)

   (b) (i) Define the cumulative distribution function (cdf) of a random variable X.

   (ii) List 3 properties of the cdf.

   (iii) Define the probability density function (pdf) for a continuous random variable.

   (iv) Show that

$$f(x) = \begin{cases} \frac{x}{2} & \text{for } x \in [0,2] \\ 0 & \text{otherwise} \end{cases}$$

   is a valid pdf.

   (v) If X is a random variable with pdf equal to $f(x)$ from part (iv), find the following probabilities:

   A. $P(X \leq 0.5)$,

   B. $P(X = 0)$,

   C. $P(\sqrt{X} > 0.5)$. (17)

   (c) Define a p-value with reference to hypothesis testing. (3)

   (d) Waiting times to speak to an operator, in minutes (rounded to integer values), for 20 randomly selected customers are:

12, 0, 6, 11, 0, 4, 9, 7, 4, 3, 1, 4, 14, 20, 2, 0, 1, 3, 1, 1

   The sample mean is 5.150 and the sample variance is 30.029. Consider modelling this data set as a realisation of i.i.d. random variables $Y_1, \ldots, Y_{20}$, each following the Geometric distribution:

$$f(y;\theta) = \theta(1-\theta)^y \qquad \text{for } y = 0, 1, 2, \ldots$$

   where $\theta \in (0,1)$ is a parameter.

   (i) Discuss whether this is a sensible model for these data

   (ii) Given that $\mathrm{E}[Y_i] = (1-\theta)/\theta$ for $i = 1, \ldots, 20$, compute the method of moments estimate of θ.

(iii) Use the fact that $\text{Var}[Y_i] = (1-\theta)/\theta^2$ to assess the fit of the model. (9)

(e) Let X follow a discrete uniform distribution with

$$X \sim \text{Unif}(1, ..., m)$$

and $Y = X + a$, where a is a constant. Find the distribution of Y. (6)

(f) A gamma distribution, Gamma(n, θ), has moment generating function $M_X(t) = \theta^n/(\theta - t)^n$. Let $X \sim \text{Gamma}(4, 3)$. Find the expectation, $\text{E}[X]$, and variance $\text{Var}[X]$. (7)

[**50**]

SECTION B

2. The joint pdf of two random variables X and Y is given by

$$f_{X,Y}(x,y) = e^{-x-y-\theta xy}\{(1+\theta x)(1+\theta y) - \theta\}$$

for $0 < x < \infty$; $0 < y < \infty$; $0 < \theta < 1$.

(a) Show that the marginal pdf of X is $f_X(x) = e^{-x}$. (13)

(b) Find the marginal pdf of Y. (3)

(c) Find the conditional pdf of $X|Y$. (4)

(d) For what value of θ will X and Y be independent? (5)

[**25**]

3. The cycle commute time between Exeter St. Davids and Streatham campus is distributed normally with unknown variance. A group of 10 commuters made the journey using a normal bike and again on the following day using a fold-up bike. The times (in minutes) of their journeys are given below.

Cyclist number	1	2	3	4	5	6	7	8	9	10
Normal bike	11	8	9	13	10	9	12	7	12	9
Fold-up bike	11	14	10	9	13	12	11	10	12	10

(a) Conduct an appropriate one-sided hypothesis test at the 10% level to test whether the fold-up bike is slower than the normal bike. State your null and alternative hypotheses, your test statistic and its null distribution, the critical region, the p-value and your conclusion. (16)

(b) Calculate a 90% one-sided confidence interval for the difference in cycle commute times between bike types and interpret your findings. (7)

(c) Imagine we had instead tested at the 5% level. What would we conclude now? (2)

[**25**]

4. At a Tombola, prizes on a table are assigned numbers and punters play by paying to randomly draw a ticket and, if their ticket number matches the number on a prize, they win that prize. At a village fair, a Tombola with a grand prize of an iPad for ticket 100 is being held. One hundred tickets, numbered 1 to 100 were used with small prizes assigned to tickets ending in a 5. Tickets ending in 0 had bottles of wine as prizes (other than iPad-winning ticket 100). All other tickets did not win prizes and tickets cost £10 each.

   A man is 4th in the queue (behind 3 other people) prior to the Tombola opening, when a thief takes a handful of tickets. It is discovered that the thief got away with 20 tickets. The staff decide to open the Tombola with a promise that if it turns out that the iPad ticket has gone, they will refund all non-prize-winning tickets at the end of the day.

   The man decides to keep his place in the queue and intends to buy one ticket.

   (a) What is the probability that the thief has the ticket that wins the iPad? (2)

   (b) What is the probability that the man will win the iPad? (2)

   (c) The first woman in the queue wins a bottle of wine and the next two people win nothing. All 3 only bought 1 ticket each. What is the probability that the thief has the iPad ticket now? (13)

   (d) After those three people have taken their tickets what is the probability that the man will get his money back after playing? (8)

   [**25**]